

Questions

Lecture – 4

1. State and explain the Ampere's law. What is the critical form of it?
2. Drive the equation of the magnetic field for a long straight current carrying wire.
3. Formulate the magnitude of the magnetic field of a long solenoid.
4. Draw a cross-sectional view of a toroid and write down the equation of the magnetic field of it.
5. Define the hall effect and hall voltage. What are the applications of it?
6. A solenoid is $1.0m$ long and $3.0cm$ in mean diameter. It has five layers of windings of 850 turns each and carries current of $5.0amp$. (a) What is B at its centre? (b) What is the magnetic flux ϕ_B for a cross-section of the solenoid at this centre? Ans: a) $2.67 \times 10^{-2}T$ b) $1.88 \times 10^{-5}wb$
7. An air-core solenoid with 2000 loops on it is $600mm$ long and has diameter of $20mm$. If a current of $5A$ is sent through it, what will be the flux density within? Ans: $2.09 \times 10^{-2}T$
8. A solenoid $0.5m$ long has 2000 turns. The magnetic induction near the centre of the solenoid is $0.08T$. What is the current in the solenoid? Ans: $15.92A$
9. A diameter of a circular coil is $31.4 \times 10^{-2}m$ and its number of turns is 400. For what amount of current flow in the coil, the magnetic field at the center of the coil will be $8 \times 10^{-10}wbm^{-2}$?
Ans: $5 \times 10^{-7}A$
10. $5A$ current is flowing through a conducting wire of length of $1m$. Calculate the magnetic field at a distance $5cm$ from the wire. Ans: $2 \times 10^{-5}T$
11. The length and diameter of a solenoid are respectively $2m$ and $2cm$. It has 10 layers. Each layer consists of 1000 turns. Determine the field intensity and flux at the center of the solenoid of $5A$ current flows in it. Ans: a) $3.14 \times 10^{-2}T$ b) $9.85 \times 10^{-6}wb$
12. The radius of the circular coil is $0.02m$ and its number of turns is 100. Calculate the magnetic field at the center of the coil if $5A$ current flows in the coil. Ans: $0.0157T$
13. A 50 turns rectangular coil has length of $0.25m$ and width of $0.2m$. If the coil carries current of $4A$ and is placed in a uniform magnetic field of $0.2T$. What magnitude of torque will act on the coil?
Ans: $2Nm$